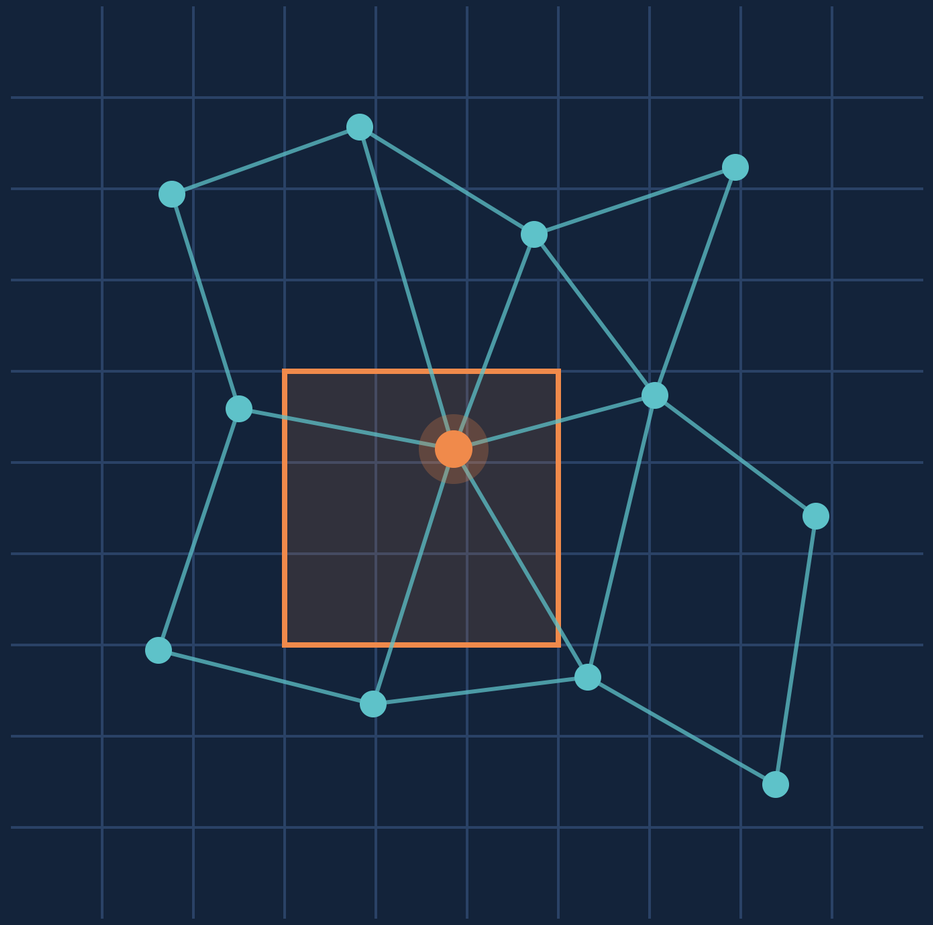
INTRODUCTION TO GEOSPATIAL AI · WEEK 1

Model architectures for GeoAI

What makes an AI model *spatial*?



Lecture 2 h · Lab this afternoon: “Same data, four architectures”

THE COURSE

Seven weeks, four arcs, one project

TOOLKIT

W1 Model architectures

You are here

EARTH FROM ABOVE

W2 Remote sensing and Earth observation

W3 Environmental monitoring

SOCIETY ON THE GROUND

W4 Big geodata mining

W5 Urban analytics

COMMUNICATING AND GOVERNING

W6 AI and cartography

W7 Ethics of GeoAI

WEEKS 8–10

Project proposal talks (W8) → project labs (W9–10) → report. Every lab is a toolkit for your project.

TODAY

More than half of today is you talking

0–10	10–30	30–50	50–60	60–80	80–110	110–120
Poll	**Roots, and why space is special**	**Paper clinic (student-led)**	**Break**	**Architecture tour**	**Architecture matchmaking and design canvas**	**Wrap**

■ Orange: you discuss ■ Teal: short input Widths are to scale (minutes)

Phones out for three live polls

Sit in groups of four for the whole session

Lab at 13:00 tests today’s ideas in code

POLL · 5 MINUTES

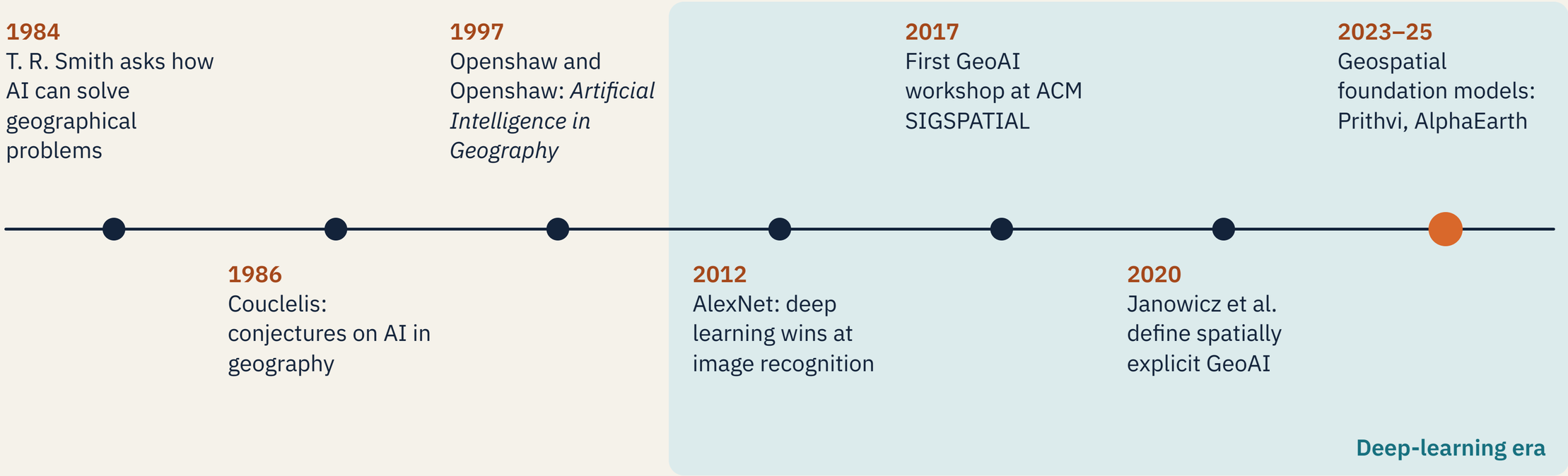
Is this GeoAI? Yes, no, or it depends

- A ride-hailing app predicts your arrival time
- A crop-type map made from satellite images
- A chatbot names the nearest late-night pharmacy
- A global 10-day weather forecast
- An email spam filter
- A house-price model that uses the postcode

Then Pick the case your table disagreed on most. What would make it *more* spatial?

ROOTS

GeoAI's questions are 40 years old; its tools are new



WHAT MAKES A MODEL SPATIAL?

Four tests of a spatially explicit model

1 · Invariance

Shuffle the locations. Do the results change?

2 · Representation

Is location (coordinates, topology, neighbours) in the data?

3 · Formulation

Does the model use distance or neighbourhood?

4 · Outcome

Do the outputs have spatial structure the inputs lacked?

A model can pass some tests and fail others: spatial explicitness is a spectrum.

PEER INSTRUCTION · VOTE, ARGUE, RE-VOTE

A CNN labels satellite tiles as “forest”, “river” or “town”. Which tests does it pass?

A

None of them

B

Representation only

C

Representation and formulation

D

All four

1 Vote alone (1 min) **2** Convince a neighbour (2 min) **3** Vote again

WHY SPACE IS SPECIAL

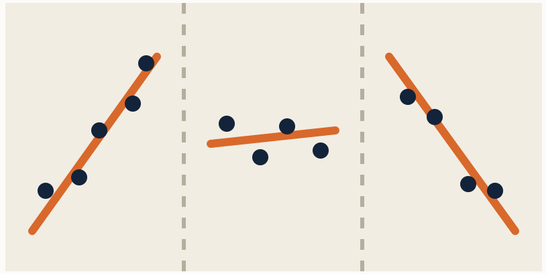
Geographic data break the “independent samples” assumption



Autocorrelation

Near things are more related than distant things.

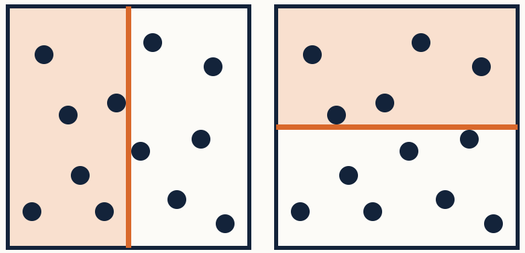
Tobler’s first law (1970)



Heterogeneity

The same relationship changes from place to place.

Often called the “second law”



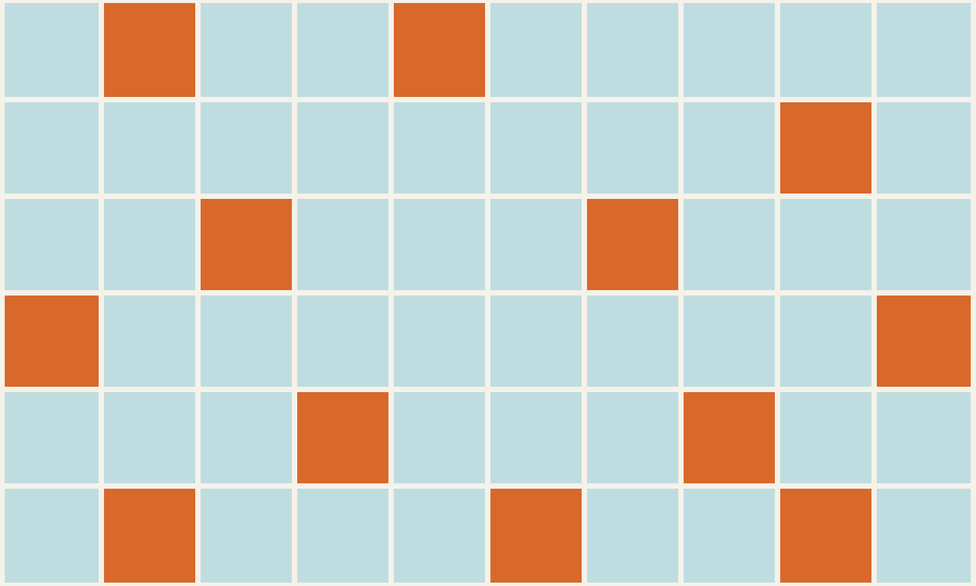
Scale and zoning

Same points, different zones, different answers.

Modifiable areal unit problem

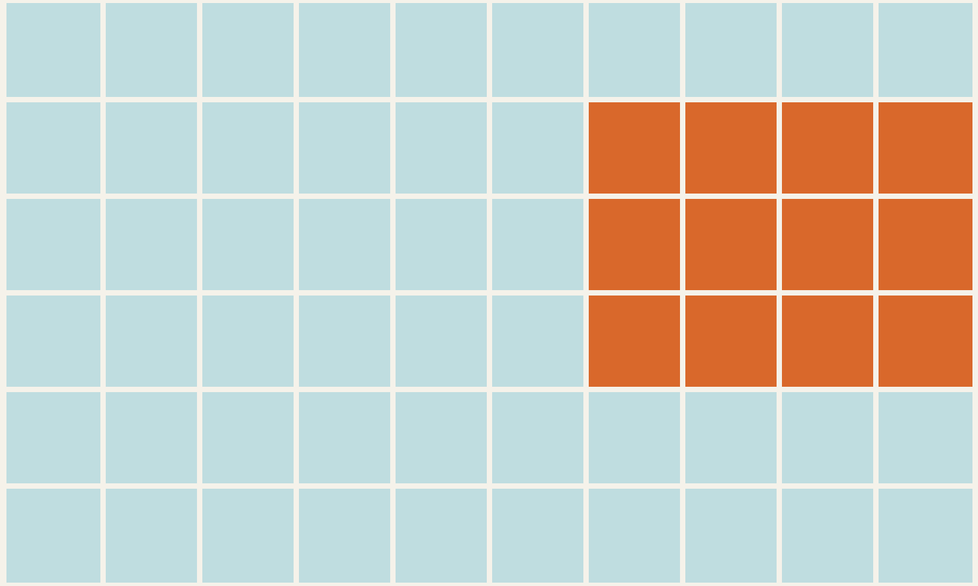
PEER INSTRUCTION · PREDICT BEFORE THE LAB

Random or spatial split: which test score is honest?



Random 5-fold

Every test cell has training neighbours. The model can interpolate.



Spatial blocks

The test region is new territory. The model must generalise.

■ test cells ■ training cells · Roberts et al. (2017)

REMEMBER THIS

A model that memorises places looks brilliant until it meets a new place.

So in this course every score comes with its split: random, or spatial.

PAPER CLINIC · 20 MINUTES

Jigsaw: become the expert, then teach your table

ROUND 1 · 8 MIN · EXPERT GROUPS

A — **Janowicz et al. (2020)**
What makes GeoAI “spatially explicit”?

B — **LeCun, Bengio and Hinton (2015)**
Which architecture suits which data, and why?

C — **Mai et al. (2022)**
How can a neural network “read” a coordinate?

ROUND 2 · 10 MIN · MIXED TABLES

A B C

Each expert teaches their paper in 3 minutes.

The table agrees one question to put to the class.

From Week 2 a student group leads this clinic and chooses the questions.

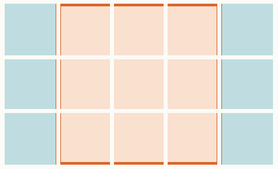
BREAK · 10 MINUTES

After the break: every architecture is a bet about the structure of your data.

Think about it over coffee: what structure does *your* favourite dataset have?

ARCHITECTURE TOUR

Every architecture is a bet about structure



GRID

CNN

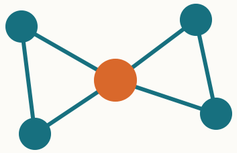
Bet: nearby pixels matter, and patterns repeat anywhere.



SEQUENCE

RNN / LSTM

Bet: the recent past shapes the present.



GRAPH

GNN

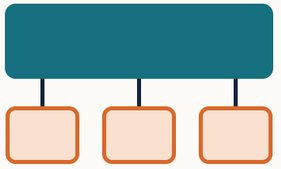
Bet: connected things influence each other.



LOCATION

Location encoder

Bet: where you are carries signal at many scales.



MANY MODALITIES

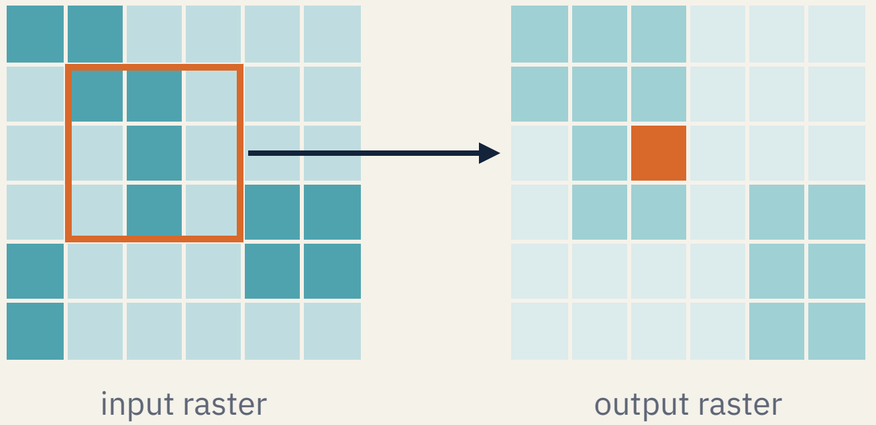
Foundation model

Bet: pretrain broadly once, adapt with few labels.

A convolution is a GIS focal operation

GIS FOCAL MEAN · YOU CHOOSE

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



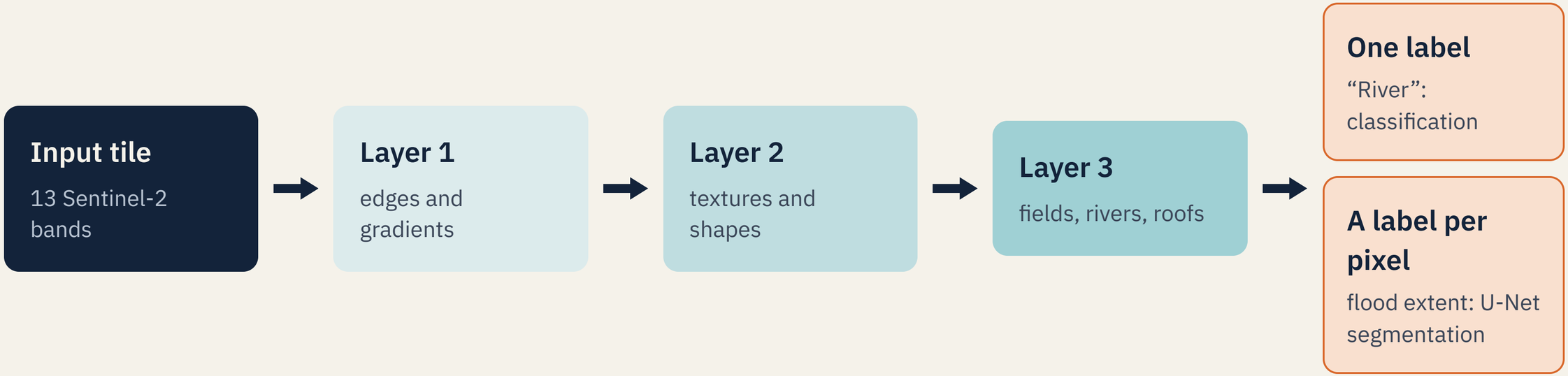
CNN FILTER · IT LEARNS

−1	0	1
−2	0	2
−1	0	1

Same sliding window. GIS picks the weights; a CNN learns them, many filters at once, layer on layer.

GRIDS · CNN

Stacked filters see ever larger patterns

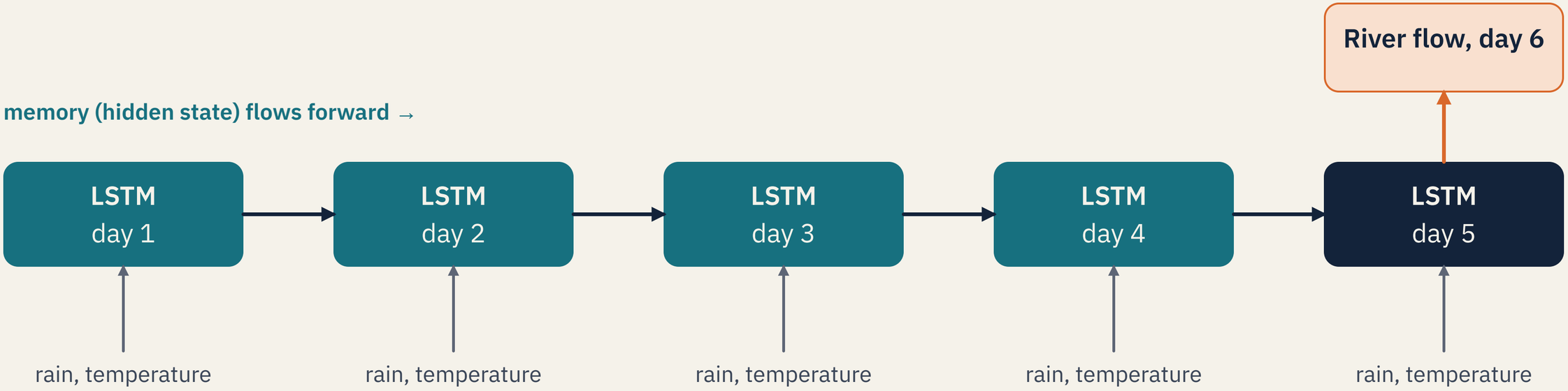


Week 2 Classify EuroSAT land-cover tiles (Helber et al., 2019)

Week 3 Map floods from radar with a U-Net (Ronneberger et al., 2015)

SEQUENCES · LSTM

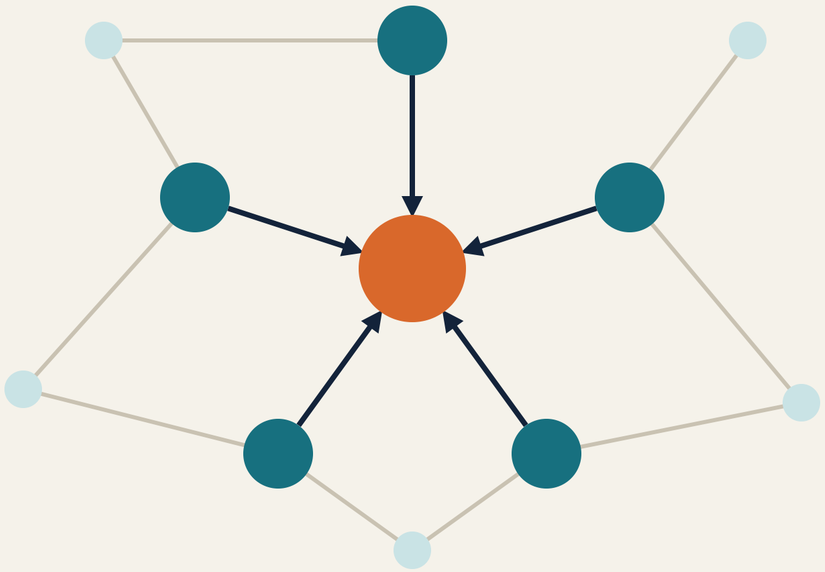
An LSTM carries a memory of the catchment through time



Week 3: LSTMs forecast river flow, even in ungauged catchments (Kratzert et al., 2019; Nearing et al., 2024).

GRAPHS · GNN

A graph neural network lets each place listen to its neighbours



ONE LAYER OF MESSAGE PASSING

me + mean of my neighbours

→ learned weights → new me

Two layers = news from two hops away. No coordinates needed: location is *who is connected to whom*.

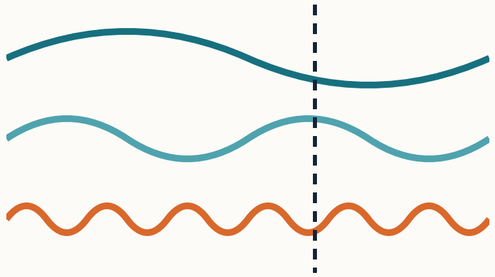
Census areas · today’s lab | Road traffic · Week 5 | Global weather mesh · GraphCast (Lam et al., 2023)

Teach a network to read a coordinate at many scales

Glasgow

(−4.25, 55.86)

longitude, latitude



sin and cos at many wavelengths: country, city, street



embedding

MLP

+ census features → house value

RAW LON/LAT

Two numbers. The network must bend them into local detail, slowly.

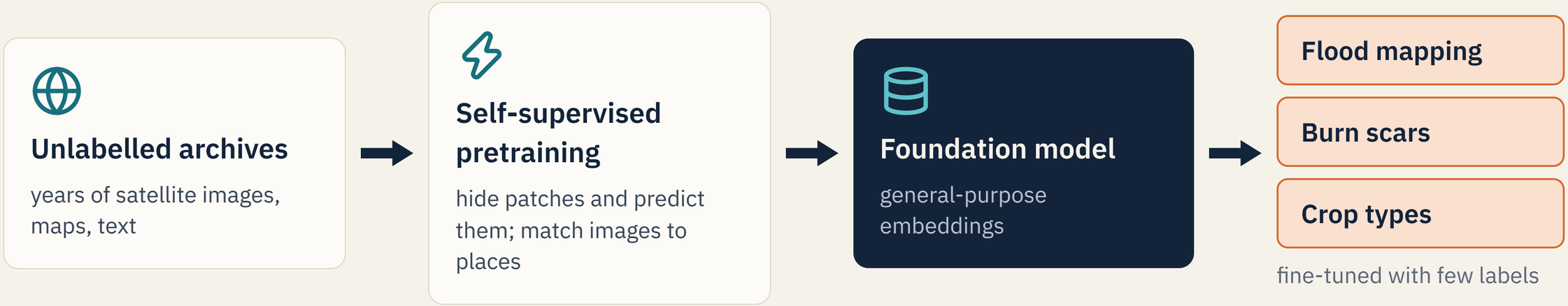
PRETRAINED ENCODERS

SatCLIP and GeoCLIP learn location embeddings from satellite images and photos.

Mai et al. (2020, 2022) · Klemmer et al. (2025) · Vivanco Cepeda et al. (2023)

MANY MODALITIES · FOUNDATION MODELS

Pretrain once on the planet, adapt with a few labels



Discuss · 2 min If every team fine-tunes the same base model, whose blind spots do we all inherit?

Examples: Prithvi-EO-2.0 (2024) · AlphaEarth (2025) · OlmoEarth (2025) · TESSERA (2025)

GROUP ACTIVITY · 12 MINUTES

Architecture matchmaking

1
Flood extent from Sentinel-1 radar images

2
Bike-share demand at 300 docking stations

3
Tomorrow’s flow at a river gauge

4
House prices from address and census data

5
Land cover from 13-band image tiles

6
Traffic speed across a city road network

7
Guess where a street photo was taken

8
Building footprints from aerial photos

Place each card on your A3 board: **which structure, which architecture?** Disagree out loud.

GROUP ACTIVITY · 18 MINUTES

The GeoAI design canvas

1

Data

What is observed, where, when, and by whom?

→

2

Structure

Grid, sequence, graph, points or several?

→

3

Model

Which bet fits that structure, and why?

→

4

Validation

Random or spatial split? What is a “new place” here?

→

5

Risk

Who could be harmed, or missed?

10′ Pick one matchmaking card and fill the A3 canvas

1′ Pitch it: one minute per table, one question from the room

ETHICS MINUTE

Your land-cover model learned from European tiles. It is now mapping Lagos.

What breaks first?

Who would notice?

Who is accountable?

LAB · 13:00 · GOOGLE COLAB

This afternoon: same data, four architectures

Random vs spatial CV

Measure the honesty gap. Was your prediction right?

Location encoding

An MLP with raw lon/lat versus multi-scale sines.

Convolution as focal filter

Prove they match, then watch a CNN learn a kernel.

Graph neural network

Message passing over neighbouring census areas.

Bring a Google account for Colab and a GitHub account: your project repository starts today.

EXIT TICKET · 3 MINUTES

Before you go

One idea from today you could explain to a friend

One thing that is still muddy

Re-vote: is a house-price model with a postcode GeoAI?

READ BEFORE WEEK 2 · EARTH OBSERVATION

Zhu et al. (2017) Deep learning in remote sensing: a comprehensive review

Helber et al. (2019) EuroSAT: a land-use and land-cover benchmark

Rolf et al. (2024) Satellite data is a distinct modality in machine learning

Annotate in the shared reader. Group 1 leads the paper clinic.

REFERENCES

Readings and sources

PRE-READINGS

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Tobler (1970) *Economic Geography* 46: 234–240

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